

Matter, Chemical Trends and Chemical Bonding

Name: _____

Instructions:

You will have the entire period to complete the test (75 minutes)

The test is made up of 4 sections:

Section A – Multiple Choice / True and False

Section B – Matching

Section C – Short Answer

Section D – Long Answer

Please **follow** all the instructions

Please **read** all questions **carefully**

A mark breakdown of the test can be found after these instructions

Good Luck!

Mark Summary

Section	Knowledge /Understanding	Thinking	Application	Communication	TOTAL
A					
B					
C					
D					
Sub-Totals	/ 31	/ 22	/ 16	/ 11	/ 80

Comments:

Section A – Multiple Choice

Please read each question carefully and then **circle** the letter of the response that you feel is the **most correct**. (Approx. 10 min for multiple choice section – 1 min/question)
(1 mark each – K/U)

1) Which of the following ions always contain more than one item?

cations

multi-valent ions

poly-atomic ions

anions

2) What is the energy change that occurs when an electron is gained by an atom in the gaseous state?

ionization energy

electron affinity

ionization

electro-negativity

3) Which is a molecular compound?

NH_3

CaCl_2

NO^{3-}

Na_2O

4) What type of bond is likely to form between the elements molybdenum, Mo and chlorine, Cl?

Non polar covalent

Polar covalent

Ionic

None of the above

5) Which of the following is true of DEET?

It is synthetically prepared

It is white and powdery solid

It will not painted surfaces

All of the above

6) Of the following attractions which is generally the weakest attractive force between particles?

Ionic bond

Covalent bond

Dipole - dipole force

London dispersion force

7) Which of the following substances is most likely to have hydrogen bonding?

Ammonia, NH_3 (g)

Methane, CH_4 (g)

Phosphine, PH_3 (g)

All of the above

8) Which property of the element decreases as you proceed from left to right across a row on the periodic table?

Atomic radius

Ionic radius

Ionization energy

Electron affinity

9) Which one of the following trends is incorrect?

a) Atomic mass always increases in a period.

b) Ionization energy always decreases in a group/family.

c) Atomic radius always increases in a group/family.

d) Number of protons always increases in a period.

10) Which one of the following is generally NOT a property of ionic compounds?

a) High melting points

b) High boiling points

c) Brittle in the solid state

d) Conducts electricity in the solid state

Section A – True/False

Please read the statements below and then indicate whether they are true or false by writing either true or false (Approximately 5 minutes for true/false section – 1 min/question). If you think the statement is false, rewrite it to make it true.

(1mark each - K/U)

Bohr's model of the atom includes neutrons located in energy levels.

The noble gases exist as individual atom and are quite reactive.

Moving down a group, ionization energies tend to decrease.

The majority of known compounds are molecular.

The correct name for the compound $(\text{NH}_3)_2\text{SO}_4$ is ammonium sulphate.

Atomic radius is measurement of the size of an atom.

There are no attractions (bonds) between two metal atoms.

Section B – Matching (1 mark each – K/U)

Match the term on the left with the most appropriate definition on the right.

1.	Polar Covalent Bond		A	The energy that is needed to remove an electron from a neutral atom.
2.	Isotopes		B	A chemical bond in which two electrons are shared by two atoms.
3.	Electron Affinity		C	An arrangement of eight electrons in the valence shell of an atom.
4.	Ionization Energy		D	A covalent bond between atoms that have significantly different electro negativities. This results in the electron pair being shared unevenly.
5.	Atomic Radius		E	A positively or negatively charged particle that results from a neutral atom or group of atoms giving up or gaining electrons
6.	Stable Octet		F	The distance from the nucleus of an atom to the approximate outer boundary of the cloud-like region of its electrons.
7.	Covalent Bond		G	Atoms of an element that are chemically similar but have different numbers of neutrons and thus, different mass numbers.
8.	Electro negativity		H	A relative measure of an atom's ability to attract shared electrons in a chemical bond.

Section C – Short Answer

Answer the following questions in the space provided. Please be sure to show complete answers. (Approx. 20 min for short answer section)

What are the three possible ways in which an atom can achieve a complete octet.

(3 marks – A)

Draw Bohr-Rutherford diagrams for atoms of the following elements: (3 marks - T/I)

Aluminum 27

Oxygen 16

Hydrogen 1

Draw Lewis diagrams for atoms of the following elements. (2 marks - T/I)

Silicon

Chlorine

An aluminum atom contains 13 protons and 14 neutrons. (3 marks - K)

What is the atomic number of Aluminum?

What is the mass number of this Aluminum?

What energy change occurs when an atom gains electron and when atom loses an electron? (2 marks - T/I)

Give an example of an electrolyte and explain why it is classified as an electrolyte.

(2 marks - T/I)

Describe the “zero-sum rule” for writing chemical formula for ionic compounds.

(2 marks - C)

Name each of the following polyatomic ions. (3 marks – K/U)



List the three different types of intermolecular forces? Which of these types are referred as Van der Waals forces? (4 marks C)

A student records the following evidence in a lab book.

Unknown substance	Pure state	Solubility in water	Solution conductivity
I	solid	high	low
II	solid	low	low
III	solid	high	none
IV	solid	high	high

Which of the substances in the table above is most likely an ionic compound?
Explain your answer. (3 marks - T/I)

For a science fair project, a student wants to design a simple device for removing certain gases from polluted air. He knows that polar molecules dissolve well in water, so he bubbles polluted air through a jug of water to remove unwanted gases. For his project, the student uses air containing the following gases: (4-marks - T/I)

- | | |
|-----------------------|-----------------------------------|
| 1. $\text{N}_{2(g)}$ | 5. $\text{CH}_{4(g)}$ |
| 2. $\text{O}_{2(g)}$ | 6. $\text{OCl}_{2(g)}$ |
| 3. $\text{HF}_{(g)}$ | 7. $\text{C}_3\text{H}_{8(g)}$ |
| 4. $\text{NH}_{3(g)}$ | 8. $\text{CH}_3\text{OCH}_{3(g)}$ |

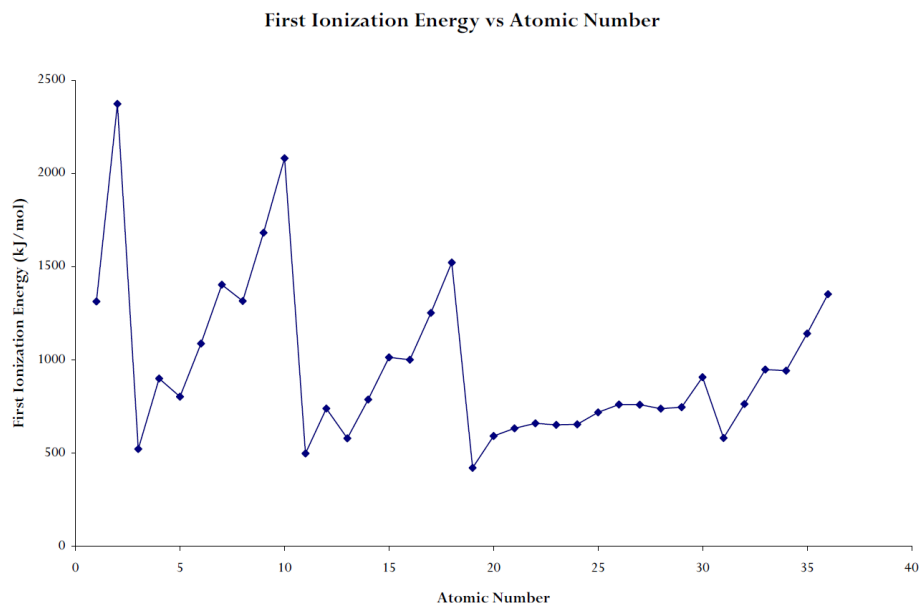
List the gases that will dissolve by writing down their corresponding numbers.

Why do you think it is important for chemists around the world to decide on a standardized naming system for chemical compounds? (2 Marks – A)

Section D – Long Answer

Read the questions carefully and then provide your answer in the spaces provided.
(Approx. 20 min for long answer section)

Examine the following graph and answer the questions below: (A)



- What is the trend (from L to R) across a period? (1 mark)
- Using the three major factors discussed in class, explain why this trend occurs. (3 marks)
- What is the trend from top to bottom in a group? (1 mark)
- Using the three major factors discussed in class, explain why this trend occurs. (3 marks)

Use the information given below to calculate the average atomic mass of element X
(3 marks - A)

Isotope	Abundance (%)	Mass (u or g/mol)
^{302}X	12.64	302.04
^{304}X	18.23	304.12
^{306}X	69.13	305.03

Determine the shape of the following molecules by drawing a Lewis structure and considering the distribution of electron pairs around the atoms. Then, determine ΔEN for the bonds. Finally, use the ΔEN and the shape to predict whether the molecule is polar or non-polar. (6 Marks – T/I)

a. H_2O

b. CCl_4

Beakers (A) and (B) are filled with substances that are liquid at room temperature. The substance in beaker (A) experiences London dispersion forces. The substance in beaker (B) experience dipole-dipole interactions. Compare and contrast the two force types. Which substance will boil at a lower temperature and why? (5 marks – C)